LECTURE 2

Claim Every $\vec{x} \in \mathbb{R}^2$ is in the form $a\begin{bmatrix}1\\0\end{bmatrix} + b\begin{bmatrix}0\\1\end{bmatrix}$ for some $a, b \in \mathbb{R}$.

Proof Given $\vec{x} \in \mathbb{R}^2$, we write

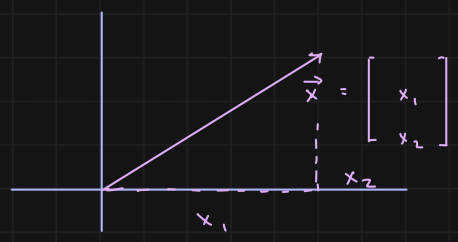
$$\vec{x} = \begin{bmatrix}x_1\\x_2\end{bmatrix} = \begin{bmatrix}x_1\\0\end{bmatrix} + \begin{bmatrix}0\\x_2\end{bmatrix} = x_1\begin{bmatrix}1\\0\end{bmatrix} + x_2\begin{bmatrix}0\\1\end{bmatrix}$$

for some $x_1, x_2 \in \mathbb{R}$.

Proposition <u>Geometry on vectors</u>

Using pythagoras, the length of $\vec{x} \in \mathbb{R}^2$ is

$$\sqrt{x_1^2 + x_2^2}$$



Def'n Let $\vec{x} \in \mathbb{R}^n$. The <u>length</u> (i.e. <u>norm</u> or <u>magnitude</u>) of $\vec{x}$ is

$$\|\vec{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

Remark If $\vec{x} = [x] \in \mathbb{R}^1$, then the norm of $\vec{x}$ is $\|\vec{x}\| = \sqrt{x_1^2} = |x_1|$

Ex

$$\left\| \begin{bmatrix}1\\0\\-1\\2\end{bmatrix} \right\| = \sqrt{1^2 + 0^2 + (-1)^2 + 2^2} = \sqrt{6}$$

Proposition <u>Properties of norm</u>

(1) If $\vec{x} \in \mathbb{R}$ and $c \in \mathbb{R}$ then $\|c\vec{x}\| = |c|\,\|\vec{x}\|$

(2) If $\vec{x} \in \mathbb{R}^n$ then $\|\vec{x}\| \geq 0$ and $\|\vec{x}\| = 0$ iff $\vec{x} = \vec{0}$.

(3) triangle inequality $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$

geometry in vectors, norm, properties of norm

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Proof

(2) We have $\|\vec{x}\| = \sqrt{x_1^2 + \dots + x_n^2} \geq 0$.

Observe that $\sqrt{x_1^2 + \dots + x_n^2} = 0$, iff $x_1^2 + \dots + x_n^2 = 0$, i.e. $x_1 = \dots = x_n = 0$, i.e.

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} = \vec{0}$$

Def'n

Let $\vec{x}, \vec{y} \in \mathbb{R}^n$. The <u>dot product</u> of $\vec{x}$ and $\vec{y}$ is $\vec{x} \cdot \vec{y} = x_1 y_1 + \dots + x_n y_n \in \mathbb{R}$.

Proposition

<u>Properties of dot product</u>

(1) symmetry $\vec{x} \cdot \vec{y} = \vec{y} \cdot \vec{x}$

(2) linearity $\vec{x} \cdot (\vec{y} \cdot \vec{z}) = \vec{x} \cdot \vec{y} + \vec{x} \cdot \vec{z}$

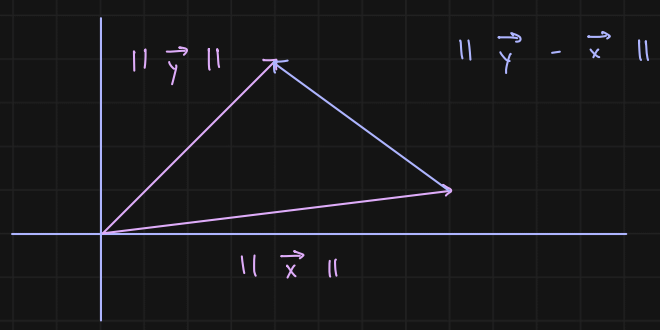
(3) $(c\vec{x}) \cdot \vec{y} = \vec{x} \cdot (c\vec{y}) = c(\vec{x} \cdot \vec{y})$

(4) non-negativity $\vec{x} \cdot \vec{x} = \|\vec{x}\|^2 = x_1^2 + \dots + x_n^2$
so $\vec{x} \cdot \vec{x} \geq 0$ and $\|\vec{x}\| = \sqrt{\vec{x} \cdot \vec{x}}$

Ex

$$\begin{bmatrix} 1 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = (1)(1) + (-2)(1) = -1$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = (1)(1) + (2)(2) + (3)(3) = 14$$



dot product, properties of dot product

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Theorem

<u>Cosine law</u>

$$\|\vec{y} - \vec{x}\|^2 = \|\vec{y}\|^2 + \|\vec{x}\|^2 - 2\|\vec{x}\|\|\vec{y}\|\cos\theta$$

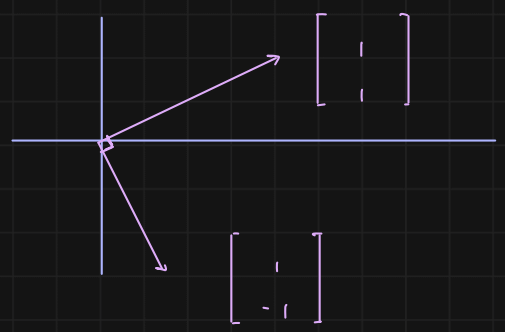
We can expand $(\vec{y} - \vec{x}) \cdot (\vec{y} - \vec{x})$ to get

$$= \vec{y} \cdot \vec{y} + \vec{x} \cdot \vec{x} - \vec{y} \cdot \vec{y} - \vec{y} \cdot \vec{x} - \vec{x} \cdot \vec{y} + \vec{x} \cdot \vec{x}$$
$$= \vec{y} \cdot \vec{y} + \vec{x} \cdot \vec{x} + 2\,\vec{x} \cdot \vec{y}$$
$$= -2\|\vec{x}\|\|\vec{y}\|\cos\theta$$

Def'n

Given $\vec{x}, \vec{y}$ in $\mathbb{R}^n$, $\vec{x}$ and $\vec{y}$ are <u>orthogonal</u>, i.e <u>perpendicular</u> iff $\vec{x} \cdot \vec{y} = \vec{0}$.

Ex



Observe that $\begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \end{bmatrix} = (1)(1) + (1)(-1) = 0$

Hence $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is orthogonal to $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$.

cosine law, perpendicularity